13 REPRODUCTION IN PLANTS

Plants, like animals, can reproduce sexually and asexually. The sexual organs of a flowering plant are its flowers, which produce pollen and ovules containing the flower's gametes. This chapter looks at both types of reproduction in flowering plants.

LEARNING OBJECTIVES

- Describe the differences between sexual and asexual reproduction in plants
- Understand that plants can reproduce asexually by natural methods (illustrated by runners) and by artificial methods (illustrated by cuttings)
- Describe the structures of an insect-pollinated and a wind-pollinated flower and explain how each is adapted for pollination
- Understand that the growth of the pollen tube followed by fertilisation leads to seed and fruit formation
- Understand how germinating seeds utilise food reserves until the seedling can carry out photosynthesis
- Investigate the conditions needed for seed germination

SEXUAL AND ASEXUAL REPRODUCTION

You have seen how animals can reproduce sexually and asexually (Chapter 9). Plants can also carry out both types of reproduction. Table 13.1 summarises some of the differences between the two methods of reproduction.

Table 13.1 Sexual and asexual reproduction compared.

Feature of the process	Sexual reproduction	Asexual reproduction
gametes produced	yes	no
fertilisation takes place	yes	no
genetic variation in offspring	yes	no
has survival value in:	changing environment	stable environment

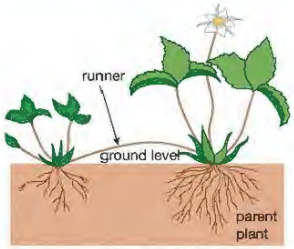
KEY POINT

As each method of asexual reproduction involves some part of the plant growing, new cells must be produced. These cells are produced by **mitosis** and so are all genetically identical. This means that all the offspring formed by asexual reproduction will also be genetically identical.

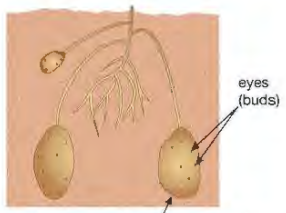
ASEXUAL REPRODUCTION IN PLANTS

There are many different methods of asexual reproduction in plants. Most involve some part of the plant growing, and then breaking away from the parent plant before developing into a new plant (Figure 13.1).

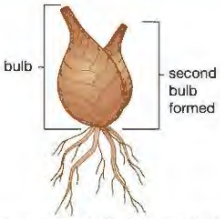
Gardeners often take advantage of the ways that plants can reproduce asexually. They use runners, bulbs and tubers to produce more plants.



A new plant is produced where the runner touches the ground.



Potato tubers form underground at the ends of branches from the main stem. Each potato can produce several new plants from the 'eyes' which are buds.



Some plants form bulbs. They are the bases of leaves which have become swollen with food. Buds in them can develop into new plants. Plants can form more than one bulb.

▲ Figure 13.1 Some methods of asexual reproduction in plants

Another type of asexual reproduction is to grow plants from **cuttings**. A piece of a plant's stem, with a few leaves attached, is cut from a healthy plant. This is planted in damp soil or compost, where it will grow roots and develop into a new plant (Figure 13.2).



▲ Figure 13.2 These geranium cuttings have started to grow roots

REMINDER

Haploid cells have only half the number of chromosomes of other body cells of that species.

SEXUAL REPRODUCTION IN PLANTS

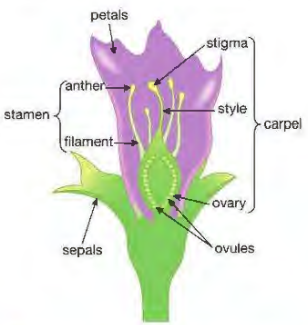
Plants produce specialised, haploid gametes in their flowers. The male gametes are contained within the **pollen grains** and the female gametes are egg cells or **ova**. Just as in animals, the male gametes must be transferred to the female gametes. This takes place through **pollination**, which is normally carried out either by wind or insects. Following pollination, fertilisation takes place and the zygote formed develops into a **seed**, which, in turn, becomes enclosed in a **fruit**.

PRODUCTION OF GAMETES AND POLLINATION

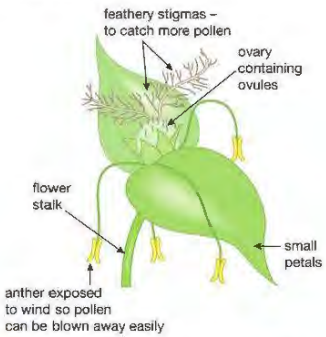
The gametes are produced by meiosis in structures in the flowers. Pollen grains are produced in the **anthers** of the **stamens**. The ova are produced in **ovules** in the **ovaries**.

In pollination, pollen grains are transferred from the anthers of a flower to the **stigma**. If this occurs within the same flower it is called **self-pollination**. If the pollen grains are transferred to a different flower, it is called **cross-pollination**. Pollination can occur by wind or by insect in either case.

Plants that are wind-pollinated produce flowers with a different structure to those of insect-pollinated flowers. These differences are related to the different methods of pollination of the flowers. Figure 13.3 shows the structure of a typical insect-pollinated flower and Figure 13.4 shows the structure of a typical wind-pollinated flower. Table 13.2 summarises the main differences between insect-pollinated flowers and wind-pollinated flowers.



▲ Figure 13.3 The main structures in an insect-pollinated flower.



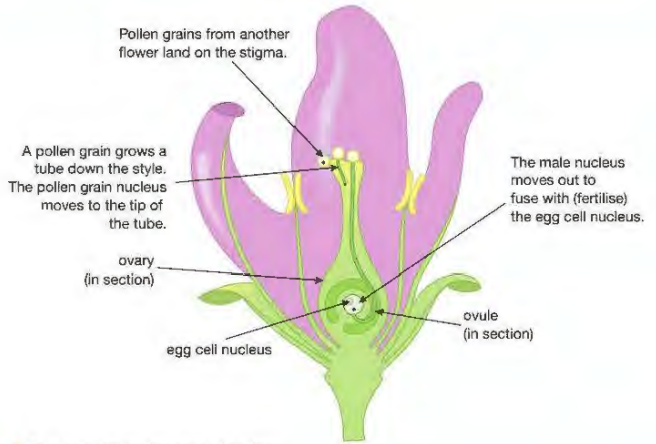
▲ Figure 13.4 The main structures in an wind-pollinated flower.

Table 13.2 Differences between insect-pollinated and wind-pollinated flowers.

Feature of flower	Type of flower	
	Insect-pollinated	Wind-pollinated
position of stamens	enclosed within flower so that insect must make contact	exposed so that wind can easily blow pollen away
position of stigma	enclosed within flower so that insect must make contact	exposed to catch pollen blowing in the wind
type of stigma	sticky so pollen grains attach from insects	feathery, to catch pollen grains blowing in the wind
size of petals	large to attract insects	small
colour of petals	brightly coloured to attract insects	not brightly coloured, usually green
nectaries	present – they produce nectar, a sweet liquid containing sugars as a 'reward' for insects	absent
pollen grains	larger, sticky grains or grains with hooks, to stick to insects' bodies	smaller, smooth, inflated grains to carry in the wind

FERTILISATION

Pollination transfers the pollen grain to the stigma. However, for fertilisation to take place, the nucleus of the pollen grain (the male gamete) must fuse with the nucleus of the ovum, which is inside an ovule in the ovary. To transfer the nucleus to the ovum, the pollen grain forms a **pollen tube**, which grows down through the tissue of the **style** and into the ovary. Here it curves around to enter the opening in an ovule. The tip of the tube dissolves and allows the pollen grain nucleus to move out of the tube and into the ovule. Here it fertilises the ovum nucleus. These events are summarised in Figure 13.5.



▲ Figure 13.5 Pollination and fertilisation

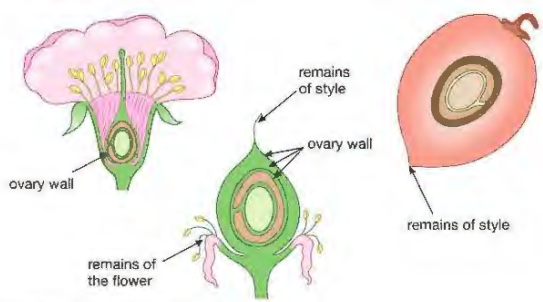
SEED AND FRUIT FORMATION

Once fertilisation has occurred, a number of changes take place in the ovule and ovary that will lead to the fertilised ovule becoming a seed and the ovary in which it is found becoming a fruit. Different flowers produce different types of fruits, but in all cases the following four changes take place.

DID YOU KNOW?
Any structure that contains seeds is a fruit. A pea pod is a fruit. The 'pod' is the fruit wall, formed from the ovary wall, and the peas are individual seeds. Each forms from a fertilised ovule.

1. The zygote develops into an embryonic plant with small root (**radicle**) and shoot (**plumule**).
2. Other contents of the ovule develop into a food store for the young plant when the seed germinates.
3. The ovule wall becomes the seed coat or **testa**.
4. The ovary wall becomes the fruit coat; this can take many forms depending on the type of fruit.

Figure 13.6 summarises these changes as they occur in the plum flower, which forms a fleshy fruit.

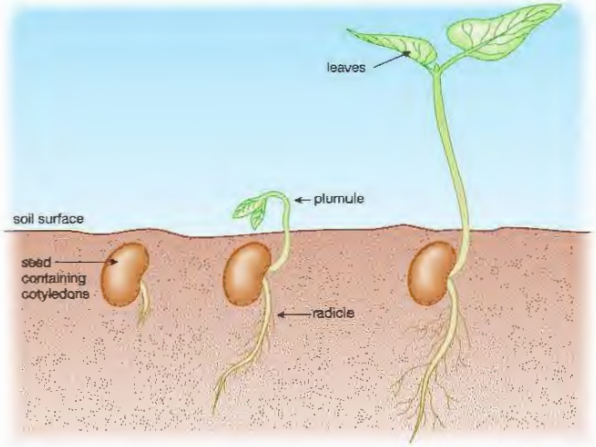


▲ Figure 13.6 How a plum fruit forms

GERMINATION

A seed contains a plant embryo, consisting of a root (radicle), shoot (plumule) and one or two seed leaves, called **cotyledons**. It also contains a food store, either in the cotyledons or another part of the seed. During germination, the food store is used up, providing the nutrients to allow the radicle and plumule to grow. The radicle grows down into the soil, where it will absorb water and mineral ions. The plumule grows upwards towards the light, where it can start the process of photosynthesis (Figure 13.7). Once the small plant (seedling) is able to photosynthesise, germination is over.

The seeds of plants such as peas or beans have two cotyledons. They are called dicotyledonous plants, or **dicots**. Seeds of grasses and other narrow-leaved plants, such as irises and orchids have only one cotyledon. They are monocotyledonous plants, or **monocots**.



▲ Figure 13.7 Germination in the broad bean. In this species, the cotyledons contain a food store. They remain below the ground when the seed germinates. In other species (e.g. the French bean), they are carried above the ground during germination.

The food store of peas or beans is present in the cotyledons. It consists mainly of starch and protein. In monocots such as maize, there is a separate food store of starch.

THE CONDITIONS NEEDED FOR GERMINATION

The growth of a new plant from a seed is called **germination**. When seeds are dispersed from the parent plant they are usually very dry, containing only about 10% water. This low water content restricts a seed's metabolism, so that it can remain alive but dormant (inactive) for a long time, sometimes for many years.

When a seed germinates, dormancy comes to an end. The seed's food store is broken down by enzymes and respired aerobically. This means that germination needs the following conditions:

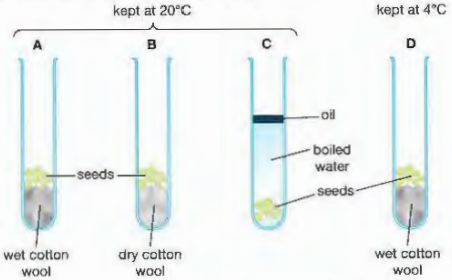
- warm temperatures, so that enzymes can act efficiently (see Chapter 1)
- water, for chemical reactions to take place in solution
- oxygen, for respiration.

Safety Note: Wash hands after handling seeds.

ACTIVITY 1

▼ PRACTICAL: INVESTIGATING THE CONDITIONS NEEDED FOR GERMINATION

Small seeds such as peas or mustard will grow on wet cotton wool in a test tube. Figure 13.A shows four tubes set up to investigate the conditions needed for the seeds to germinate.



▲ Figure 13.8 Apparatus to investigate the conditions needed for germination

The water in tube C has been boiled to remove dissolved oxygen. The layer of oil (e.g. cooking oil) keeps out oxygen from the air.

- Tube A: seeds on wet cotton wool, maintained at 20 °C (room temperature)
- Tube B: seeds on dry cotton wool, maintained at 20 °C
- Tube C: seeds in boiled water, with a thin surface layer of oil; maintained at 20 °C
- Tube D: seeds on wet cotton wool, placed in a refrigerator at 4 °C

After a few days the seeds in the control tube (A) will start to germinate. There will be no germination in tubes B or C. The seeds in tube D may eventually start to germinate, but much more slowly than in tube A.

What do the results tell you about the conditions needed for germination? Do you have any criticisms of this method? (Hint – are all the variables fully controlled?)

CHAPTER QUESTIONS

More questions on reproduction in plants can be found at the end of Unit 3 on page 182.

SKILLS CRITICAL THINKING

1 Which of the following features are shown by wind-pollinated flowers?

A large petals, a scent and sticky pollen grains
B small petals, no scent and light pollen grains
C large petals, no scent and light pollen grains
D small petals, a scent and sticky pollen grains

2 During pollination, between which of the following structures are pollen grains transferred?

A from anther to stigma **B** from anther to ovary
C from stigma to style **D** from stigma to ovary

SKILLS CRITICAL THINKING (4)

3 Following fertilisation, which of the following takes place?

A The ovule becomes the fruit and the ovary becomes the seed
B The carpel becomes the fruit and the anther becomes the seed
C The anther becomes the fruit and the carpel becomes the seed
D The ovary becomes the fruit and the ovule becomes the seed

(5)

4 Read the three statements below.

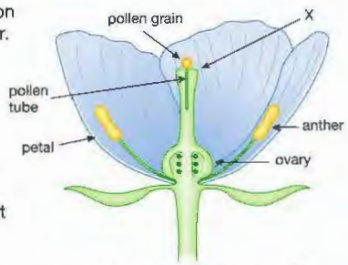
1. Uptake of water 2. Uptake of oxygen 3. Increase in metabolism

Which of these events are associated with seed germination?

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 1, 2 and 3

SKILLS ANALYSIS (4)

5 The diagram below shows a section through an insect-pollinated flower. Pollination happens when pollen grains land on part X.



a Name part X.

SKILLS REASONING

b How are insects attracted to a flower like this? Give two ways.

SKILLS INTERPRETATION (5)

c Copy the diagram and extend the pollen tube to show where it would go when fully grown.

SKILLS CRITICAL THINKING (6)

6 A student investigated whether temperature affected the germination of seeds. He set up three boiling tubes, each containing 10 seeds placed on wet cotton wool.

- Tube A was kept at room temperature (20 °C) in the light.
- Tube B was placed in a fridge at 4 °C.
- Tube C was placed in an incubator at 30 °C.

The student monitored the seeds for 3 days, making sure that the cotton wool remained damp. At the end of this time he observed the results and measured the height of any seedlings that had germinated. His results are shown in the table.

Tube	Height of any seedlings that germinated / cm	% seeds germinated	Average height of seeds that germinated / cm
A	2.3, 2.7, 1.9, 2.5, 2.6, 3.1, 1.9, 2.2, 2.6, 3.2	100	2.5
B	0.3, 0.5 (the other seeds did not germinate)	20	0.4
C	3.4, 4.5, 2.5, 3.7, 2.8, 4.4, 4.3, 2.9, 2.1, 3.7	100	

a State the independent variable and the dependent variable in the student's experiment.

SKILLS PROBLEM SOLVING (5)

b Calculate the average height of the seedlings that germinated in tube C.

SKILLS DECISION MAKING

c Summarise the conclusions that can be drawn from this experiment.

SKILLS REASONING (6)

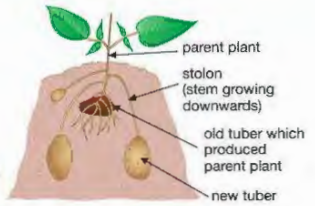
d Why does temperature affect the germination of seeds?

SKILLS EXECUTIVE FUNCTION (5)

e Another student suggested that the experiment was not properly controlled. Explain what she meant, and suggest how the experiment could be modified.

SKILLS REASONING (5)

7 The diagram shows a potato plant producing new tubers (potatoes). Buds on the parent plant grow into stems that grow downwards, called stolons. The ends of each stolon develop into a new tuber.



a Give two pieces of evidence which show that this is an asexual method of reproduction.

(7)

b Explain why all the new tubers will be genetically identical.

(6)

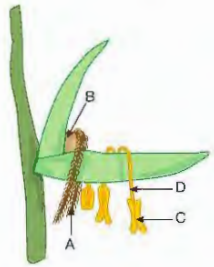
c Even though the tubers are genetically identical, the plants that grow from them may not be the same height. Explain why.

(7)

d Why do wild plants need to reproduce sexually as well as asexually?

SKILLS CRITICAL THINKING (4)

8 The drawing shows a wind-pollinated flower.



a Name the structures labelled A, B, C and D.

b Give *three* pieces of evidence *visible in the diagram*, which show that this flower is wind-pollinated.

(5)

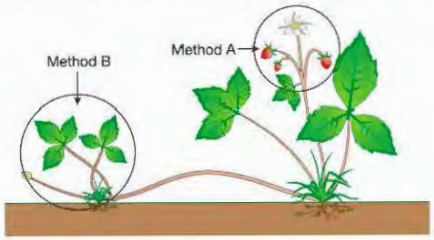
c Describe how fertilisation takes place once a flower has been pollinated.

(4)

d Describe *four* ways in which you would expect an insect-pollinated flower to be different from the flower shown.

SKILLS REASONING (7)

9 The drawing shows a strawberry plant reproducing in two ways.



a Which of the two methods of reproduction shown will result in offspring that show genetic variation? Explain your answer.

(5)

b Is the strawberry flower likely to be wind-pollinated or insect-pollinated? Give reasons for your answer.

(7)

10 All banana plants are reproduced asexually. Biologists are concerned for their future, as a new strain of fungus has appeared which is killing all the banana plants in some plantations.

a Explain why the fungus is able to kill all the banana plants in some plantations.

b Explain why this would be less likely to happen if banana plants reproduced sexually.

c Describe the benefits of reproducing banana plants asexually.